Lab Report

Computer Architecture Laboratory
Assignment 4

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1 Results

Program	Cycle Count	# times OF stage was stalled	# wrong path instructions
even-odd.asm	14	5	0
descending.asm	459	90	88
fibonacci.asm	128	31	16
palindrome.asm	93	34	7
prime.asm	45	8	5

2 Observations

The number of cycles taken to execute a program has increased compared to the single cycle implementation. This is an expected consequence of implementing pipelining in our processor as it increases the CPI, for a single cycle processor the CPI is 1, however for a processor with pipelining which encounters hazards the CPI would be greater than 1. While there may be an increase in CPI, there is still an overall gain in performance as the clock cycle time has been reduced to just the stage which takes the longest time, in case of a Single cycle processor the clock cycle time is the sum of time taken by all the stages.